Practical NO.01

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Roll No. -

Title - Write a program for pre-processing of a text document such as stop word removal, stemming.

In [1]:
```
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords')
print(stopwords.words('english'))
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am',
'an', 'and', 'any', 'are', 'aren', "aren't", 'as', 'at', 'be', 'because',
'been', 'before', 'being', 'below', 'between', 'both', 'but', 'by', 'ca
n', 'couldn', "couldn't", 'd', 'did', 'didn', "didn't", 'do', 'does', 'do
esn', "doesn't", 'doing', 'don', "don't", 'down', 'during', 'each', 'fe
w', 'for', 'from', 'further', 'had', 'hadn', "hadn't", 'has', 'hasn', "ha
sn't", 'have', 'haven', "haven't", 'having', 'he', "he'd", "he'll", 'he
r', 'here', 'hers', 'herself', "he's", 'him', 'himself', 'his', 'how',
'i', "i'd", 'if', "i'll", "i'm", 'in', 'into', 'is', 'isn', "isn't", 'i
t', "it'd", "it'll", "it's", 'its', 'itself', "i've", 'just', 'll', 'm',
'ma', 'me', 'mightn', "mightn't", 'more', 'most', 'mustn', "mustn't", 'm
y', 'myself', 'needn', "needn't", 'no', 'nor', 'not', 'now', 'o', 'of',
'off', 'on', 'once', 'only', 'or', 'other', 'our', 'ours', 'ourselves',
'out', 'over', 'own', 're', 's', 'same', 'shan', "shan't", 'she', "sh
e'd", "she'll", "she's", 'should', 'shouldn', "shouldn't", "should've",
'so', 'some', 'such', 't', 'than', 'that', "that'll", 'the', 'their', 'th
eirs', 'them', 'themselves', 'then', 'there', 'these', 'they', "they'd",
"they'll", "they're", "they've", 'this', 'those', 'through', 'to', 'too',
'under', 'until', 'up', 've', 'very', 'was', 'wasn', "wasn't", 'we', "w
e'd", "we'll", "we're", 'were', 'weren', "weren't", "we've", 'what', 'whe
n', 'where', 'which', 'while', 'who', 'whom', 'why', 'will', 'with', 'wo
n', "won't", 'wouldn', "wouldn't", 'y', 'you', "you'd", "you'll", 'your',
"you're", 'yours', 'yourself', 'yourselves', "you've"]
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\muthi\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

In [3]:
```
import nltk
nltk.download('punkt_tab')

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

example_sent = """This is a sample sentence,
showing off the stop words filtration."""
stop_words = set(stopwords.words('english'))
word_tokens = word_tokenize(example_sent)
# converts the words in word_tokens to lower case and then checks whether
#they are present in stop_words or not
filtered_sentence = [w for w in word_tokens if not w.lower() in stop_word
```

```
#with no lower case conversion
filtered_sentence = []
for w in word_tokens:
    if w not in stop_words:
        filtered_sentence.append(w)
print(word_tokens)
print(filtered_sentence)
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\muthi\AppData\Roaming\nltk_data...
['This', 'is', 'a', 'sample', 'sentence', ',', 'showing', 'off', 'the',
'stop', 'words', 'filtration', '.']
['This', 'sample', 'sentence', ',', 'showing', 'stop', 'words', 'filtrati
on', '.']
[nltk_data]   Unzipping tokenizers\punkt_tab.zip.
```

In [5]:
```
with open("text.txt", "w") as f:
    f.write("This is a sample text file for information retrieval laborat

import io
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
# word_tokenize accepts
# a string as an input, not a file.
stop_words = set(stopwords.words('english'))
file1 = open("text.txt")
# Use this to read file content as a stream:
line = file1.read()
words = line.split()
for r in words:
    if not r in stop_words:
        appendFile = open('filteredtext.txt','a')
        appendFile.write(" "+r)
        appendFile.close()
```

In [6]:
```
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize
ps = PorterStemmer()
# choose some words to be stemmed
words = ["program", "programs", "programmer", "programming", "programmers
for w in words:
    print(w, " : ", ps.stem(w))
```

```
program  :  program
programs  :  program
programmer  :  programm
programming  :  program
programmers  :  programm
```

In [7]:
```
# importing modules
from nltk.stem import PorterStemmer
from nltk.tokenize import word_tokenize
ps = PorterStemmer()
sentence = "Programmers program with programming languages"
```

```
words = word_tokenize(sentence)
for w in words:
    print(w, " : ", ps.stem(w))
```

```
Programmers  :  programm
program  :  program
with  :  with
programming  :  program
languages  :  languag
```